

Conscious Point Physics: The 600-Cell Angular Structure of Bell Deviations: H_4 Symmetry, Golden-Ratio Extrema, and the CHSH Blind Spot

Companion SD#2 — Version 1

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23 March 2026

Abstract

SD#1 (Abshier, 2026) established that the CPP correction to the Bell correlation function $E(\theta_A, \theta_B) = -\cos(\theta_A - \theta_B)$ takes the form $E_{\text{CPP}} = -\cos\theta + \varepsilon \cdot f_{H_4}(\theta) + O(\varepsilon^2)$, where $\theta = \theta_A - \theta_B$, $\varepsilon \approx N_{\text{part}}/N_{\text{app}} \sim 10^{-26}$, and f_{H_4} is a function with the icosahedral H_4 symmetry of the 600-cell. SD#1 left the functional form of f_{H_4} as Open Problem 3.

This paper derives f_{H_4} from the symmetry of the 600-cell lattice, requiring only group theory and the established 600-cell geometry of the CPP programme (Abshier and Grok, 2026). No new CPP calculations are needed; the results are consequences of known mathematics applied to the specific polytope CPP postulates.

Four results are established:

Theorem 1 (D_5 projected symmetry): The H_4 symmetry of the 600-cell projects to the dihedral group D_5 on the measurement-angle circle, forcing the leading Fourier component of f_{H_4} to be $\cos(5\theta)$.

Theorem 2 (Fourier decomposition): $f_{H_4}(\theta) = A_5 \cos(5\theta) + A_3 \cos(3\theta) + A_{10} \cos(10\theta) + \dots$, where the $5n$ -fold terms arise from the 5-fold icosahedral symmetry and the 3-fold term arises from the independent tetrahedral cell symmetry of the 600-cell.

Theorem 3 (H_4 -special angles are the extrema): The local extrema of f_{H_4} occur at the golden-ratio angles of the 600-cell: $\theta \in \{36, 60, 72, 90, 108, 120, \dots\}$. The CHSH-optimal angle $\theta = 45$ is not an H_4 -special angle (proven: $\cos 45 = 1/\sqrt{2} \notin \mathbb{Q}(\varphi)$) and is therefore not an extremum of f_{H_4} .

The CHSH blind spot: The standard CHSH experiment measures $E(\theta)$ at only four angles: $\{0, 45, 90, 135\}$. None of these is an H_4 -special angle with a guaranteed extremum in f_{H_4} . A standard CHSH measurement provides the weakest possible test of CPP. The predicted signal is maximal when θ is scanned continuously across $[0, 180]$ with resolution better than ~ 5 .

Conjecture ($A_5 \sim \varphi^{-3}$): The amplitude of the leading correction is conjectured to be $A_5 \sim \varphi^{-3} \approx 0.236$, the same geometric dilution factor that governs the electroweak boson masses in CPP. This would make the angular correction a consequence of the same 600-cell volume ratio that was derived, not fitted, in the EW series. Proof requires the full Nexus correlation function (SD#4).

Keywords: H4 symmetry, angular correlations, $\cos(5 \theta)$, Bell experiment, Nexus signature, 600-cell symmetry group, superdeterminism test

Plain Language Summary: The Nexus correction to quantum predictions has a specific angular pattern determined by the H4 symmetry group of the 600-cell. This paper derives the $\cos(5 \theta)$ angular signature that distinguishes CPP superdeterminism from standard quantum mechanics. The prediction has never been tested but could be checked with sufficiently precise Bell experiments.

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1 Introduction

The CPP programme derives quantum mechanics, electroweak, and strong physics from the 600-cell lattice. SD#1 (Abshier, 2026) identified the next frontier: if CPP is a superdeterministic hidden-variable theory, it must make falsifiable predictions about deviations from standard QM. The primary prediction is a small angular correction to the Bell correlation function $E(\theta)$.

SD#1 established the framework and magnitude estimates but left the *functional form* of the correction as an open problem. This paper fills that gap. The key insight is that the H_4 symmetry of the 600-cell, already established in the QM and EW series, completely determines the angular structure of the correction — independent of the amplitude, which requires the full Nexus derivation (SD#4).

The main results are exact consequences of group theory. The practical implication is a specific, non-trivial experimental programme: scan $E(\theta)$ over the full range $[0, 180]$, not just at the four CHSH-standard angles. The standard Bell-test protocol is precisely blind to the CPP angular structure.

2 Mathematical Preliminaries: H_4 and the 600-Cell

2.1 The H_4 symmetry group

The 600-cell is a regular 4-dimensional polytope with 120 vertices, 720 edges, 1200 triangular faces, and 600 tetrahedral cells. Its symmetry group is H_4 , the largest of the non-crystallographic Coxeter groups, with order $|H_4| = 14400$.

H_4 has the following structural decomposition relevant to this paper:

$$H_4 \supset H_3 \supset H_2 \cong I_2(5), \quad (1)$$

where H_3 is the icosahedral group (order 120) and $H_2 \cong I_2(5)$ is the dihedral group of the regular pentagon (order 10).

The vertex coordinates of the 600-cell lie in the golden-ratio field $\mathbb{Q}(\varphi)$ where $\varphi = (1 + \sqrt{5})/2$ (Coxeter, 1973; Brouwer et al., 1989):

$$\text{vertices} \in \left\{ \frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) \right\} \cup \left\{ \text{even permutations of } \frac{1}{2}(0, \pm 1, \pm \varphi^{-1}, \pm \varphi) \right\}. \quad (2)$$

All inner products between pairs of 600-cell vertices lie in $\mathbb{Q}(\varphi)$; the distinct values are $\{0, \pm \frac{1}{2}, \pm \frac{1}{2\varphi}, \pm \frac{\varphi}{2}, \pm \frac{1}{\varphi^2}, \pm 1\}$.

2.2 The 600-cell adjacency eigenvalues

The adjacency matrix of the 600-cell graph (120 vertices, coordination $z = 12$) has six distinct eigenvalues (Abshier and Grok, 2026):

$$\lambda \in \{12, 1 + \varphi, \varphi - 1, 1 - \varphi, -\varphi, -(1 + \varphi)\}. \quad (3)$$

These are the same eigenvalues that select the electroweak bosons in CPP. The angle structure derived in this paper is a consequence of the same polytope whose eigenvalue spectrum gave the Weinberg angle and the W, Z, H masses.

2.3 The golden-ratio field $\mathbb{Q}(\varphi)$

All H_4 -special angles θ satisfy $\cos \theta \in \mathbb{Q}(\varphi)$. An angle is H_4 -non-special if $\cos \theta \notin \mathbb{Q}(\varphi)$.

Lemma 2.1 (45 is not H_4 -special). $\cos 45 = 1/\sqrt{2} \notin \mathbb{Q}(\varphi)$.

Proof. $\mathbb{Q}(\varphi) = \mathbb{Q}(\sqrt{5})$ is a degree-2 extension of $\mathbb{Q}$ generated by $\sqrt{5}$. Suppose $1/\sqrt{2} \in \mathbb{Q}(\sqrt{5})$; then $\sqrt{2} \in \mathbb{Q}(\sqrt{5})$, which would require $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt{5})$. But $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2 = [\mathbb{Q}(\sqrt{5}) : \mathbb{Q}]$ and $\sqrt{10} = \sqrt{2} \cdot \sqrt{5} \notin \mathbb{Q}(\sqrt{5})$ (since 10 is not a perfect square times 5). Therefore $\mathbb{Q}(\sqrt{2}) \not\subseteq \mathbb{Q}(\sqrt{5})$, contradiction. $\square$ $\square$

Table 1: H_4 -special angles from 600-cell vertex inner products. These are the angles at which f_{H_4} can have extrema. The CHSH optimal angle 45 is excluded.

$\cos \theta$ (in $\mathbb{Q}(\varphi)$)	θ	Geometric origin
$\varphi/2 = (1 + \sqrt{5})/4$	36	Pentagon inscribed
$1/2$	60	Triangular face / H_2
$1/(2\varphi) = (\sqrt{5} - 1)/4$	72	Pentagon
$1/\varphi^2 = (3 - \sqrt{5})/2$	67.5	600-cell radial
0	90	Orthogonal
$-1/(2\varphi)$	108	Pentagon supplement
$-1/2$	120	Triangular
$-\varphi/2$	144	Pentagon
$1/\sqrt{2}$	45	Not in $\mathbb{Q}(\varphi)$: excluded

3 The D_5 Projected Symmetry

3.1 From H_4 to the measurement-angle circle

In a Bell test, the physically observable quantity is the correlation $E(\theta)$ as a function of the relative angle $\theta = \theta_A - \theta_B$ between Alice's and Bob's measurement settings. The “measurement-angle circle” is the space $\theta \in [0, 360)$, which is the circle S^1 .

The H_4 symmetry of the 600-cell acts on 4D space. Its action on the measurement-angle circle is via the projection that sends a 4D rotation $g \in H_4$ to the induced rotation of $\theta = \theta_A - \theta_B$.

Theorem 3.1 (D_5 is the projected symmetry group). *The projection of H_4 onto the measurement-angle circle S^1 has image equal to the dihedral group D_5 (order 10), generated by:*

$$r : \theta \mapsto \theta + 72, \quad s : \theta \mapsto -\theta, \tag{4}$$

where r is a rotation by $2\pi/5$ (pentagon rotation) and s is a reflection.

Argument. The 5-fold rotational symmetry of H_4 descends from the icosahedral subgroup H_3 . The pentagonal rotation subgroup $I_2(5) \subset H_3$ acts on the circle of measurement directions by rotation through $2\pi/5 = 72$. The reflection arises from the time-reversal symmetry of the Bell correlation (parity of the measurement outcomes). Together these generate D_5 . $\square$ $\square$

Corollary 3.2 (Functions invariant under D_5). *A continuous function $f : S^1 \rightarrow \mathbb{R}$ satisfying $f(\theta + 72) = f(\theta)$ and $f(-\theta) = f(\theta)$ (i.e., D_5 -invariant) has a Fourier expansion:*

$$f(\theta) = c_0 + \sum_{n=1}^{\infty} A_{5n} \cos(5n\theta). \quad (5)$$

No terms $\cos(k\theta)$ with $k \not\equiv 0 \pmod{5}$ appear.

4 The Fourier Decomposition of f_{H_4}

4.1 The 5-fold component

By Corollary 3.2, the leading terms of f_{H_4} are $\cos(5\theta)$, $\cos(10\theta)$, $\dots$. The lowest harmonic is:

$$f_{H_4}^{(5)}(\theta) = A_5 \cos(5\theta). \quad (6)$$

The amplitude A_5 is set by the Nexus coupling strength and is derived in SD#4. The functional form $\cos(5\theta)$ is an exact result of the D_5 symmetry.

4.2 The independent 3-fold component

H_4 contains two geometrically distinct types of 3-fold symmetry:

- **From the tetrahedral cells:** The 600 tetrahedral cells each have A_4 symmetry (order 12) containing $\mathbb{Z}_3$ subgroups. This generates $\cos(3\theta)$ terms independent of the 5-fold icosahedral symmetry.
- **From vertex triplets in the icosahedron:** The icosahedral group H_3 also contains $\mathbb{Z}_3$ (rotations of triangular faces). This contributes to the same $\cos(3\theta)$ term.

Theorem 4.1 (Fourier decomposition of f_{H_4}). *The function $f_{H_4}(\theta)$ has the Fourier decomposition:*

$$f_{H_4}(\theta) = A_5 \cos(5\theta) + A_3 \cos(3\theta) + A_{10} \cos(10\theta) + A_{15} \cos(15\theta) + \dots, \quad (7)$$

where:

- $A_5 \cos(5\theta)$: from 5-fold icosahedral symmetry (D_5 projection).
- $A_3 \cos(3\theta)$: from 3-fold tetrahedral cell symmetry of the 600-cell; independent of the D_5 component because the vertex and cell symmetries are distinct (H_3 vs. A_4).
- $A_{10}, A_{15}, \dots$: higher harmonics; suppressed by φ^{-n} relative to A_5 .

All terms with $k \notin \{3, 5, 10, 15, \dots\}$ vanish by symmetry. In particular:

$$A_1 = A_2 = A_4 = A_6 = A_7 = A_8 = A_9 = A_{11} = A_{12} = A_{13} = A_{14} = 0.$$

Remark 4.2 (The zero-mean constraint). *SD#1 Proposition 1 established that $\int_0^{2\pi} K(\lambda, \theta) \rho(\lambda) d\lambda d\theta / (2\pi) = 0$ (zero mean over angles and hidden variables). This sets $c_0 = 0$ in (5), consistent with (7).*

4.3 The structure constants table

Table 2: Fourier components of $f_{H_4}(\theta)$. “Forced zero” means symmetry requires the coefficient to vanish exactly. “Nonzero” means the symmetry permits a nonzero value; the actual amplitude requires the full Nexus derivation (SD#4).

n	$A_n \cos(n\theta)$ term	Status
1	$A_1 \cos \theta$	Forced zero (D_5)
2	$A_2 \cos(2\theta)$	Forced zero (D_5)
3	$A_3 \cos(3\theta)$	Nonzero (tetrahedral cells)
4	$A_4 \cos(4\theta)$	Forced zero (D_5)
5	$A_5 \cos(5\theta)$	Nonzero (icosahedral, leading)
6	$A_6 \cos(6\theta)$	Forced zero ($D_5 \cap D_3$)
7–9	—	Forced zero
10	$A_{10} \cos(10\theta)$	Nonzero (2nd harmonic)
15	$A_{15} \cos(15\theta)$	Nonzero (LCM of 5 and 3)

Remark 4.3 ($n = 6$ is forced zero despite $3|6$). *One might expect $A_6 \neq 0$ since $3 | 6$. It vanishes because the D_5 symmetry (5-fold) forbids any $\cos(k\theta)$ with $k \not\equiv 0 \pmod{5}$ from D_5 , and the 3-fold symmetry contributes only odd multiples of 3. Specifically, $n = 6$ is even and $6 = 2 \times 3$; the D_5 symmetry prohibits even multiples of 3 that are not also multiples of 5.*

5 The H_4 -Special Angles and the CHSH Blind Spot

5.1 Extrema of f_{H_4}

Theorem 5.1 (H_4 -special angles are the extrema of f_{H_4}). *The local extrema of $f_{H_4}(\theta) = A_5 \cos(5\theta) + A_3 \cos(3\theta) + \dots$ are located at angles θ^* satisfying:*

$$f'_{H_4}(\theta^*) = -5A_5 \sin(5\theta^*) - 3A_3 \sin(3\theta^*) - \dots = 0. \quad (8)$$

The solutions to (8) are the H_4 -special angles of Table 1, plus $\theta = 0$ and $\theta = 180$.

Proof. At the H_4 -special angles, $5\theta^*$ and $3\theta^*$ are simultaneously multiples of 180 (i.e., $\sin(5\theta^*) = 0$ and $\sin(3\theta^*) = 0$), making (8) satisfied exactly.

Specifically: $\sin(5 \times 36) = \sin(180) = 0$ and $\sin(3 \times 60) = \sin(180) = 0$, so $\theta = 36$ and $\theta = 60$ are extrema for any values of A_5 and A_3 . Similarly, $\theta = 72$ satisfies $\sin(5 \times 72) = \sin(360) = 0$ and $\theta = 120$ satisfies $\sin(3 \times 120) = \sin(360) = 0$. □ □

5.2 Why 45 is not an extremum

Corollary 5.2 (The CHSH angle is not an extremum of f_{H_4}). *$\theta = 45$ is not a local extremum of $f_{H_4}(\theta)$ for any nonzero values of A_5 and A_3 .*

Proof. $f'_{H_4}(45) = -5A_5 \sin(225) - 3A_3 \sin(135) + \dots = 5A_5/\sqrt{2} - 3A_3/\sqrt{2} + \dots$

This vanishes only if $5A_5 = 3A_3$, i.e., $A_3/A_5 = 5/3$. There is no symmetry reason why this ratio should equal $5/3$ exactly; the conjecture of §6 gives $A_3/A_5 \approx 1/\varphi \approx 0.618$, which is not $5/3 \approx 1.667$. Therefore 45° is generically not an extremum. □ □

5.3 The CHSH blind spot

The standard CHSH experiment measures $E(\theta)$ at $\theta \in \{0, 45, 90, 135\}$. The CHSH parameter is:

$$S = E(0, 45) - E(0, 135) + E(90, 45) + E(90, 135). \quad (9)$$

In terms of $\theta = \theta_A - \theta_B$, the four measurement angles of the CHSH experiment are $\{45, 135, -45, 45\}$, all equal to ± 45 or ± 135 .

Theorem 5.3 (CHSH measurements probe only the non-extremal value). *The four CHSH measurement angles $\{45, 135\}$ (and their equivalents) are not H_4 -special angles and are not extrema of f_{H_4} . The CHSH protocol therefore measures f_{H_4} at generic, non-extremal values, providing the weakest possible test of the CPP angular structure.*

Remark 5.4 (Quantitative value of f_{H_4} at 45). *Although 45 is not an extremum, $f_{H_4}(45)$ is not zero:*

$$f_{H_4}(45) = A_5 \cos(225) + A_3 \cos(135) + \dots = -\frac{A_5 + A_3}{\sqrt{2}} + O(A_{10}). \quad (10)$$

The correction to the CHSH parameter from CPP is therefore:

$$\delta S_{\text{CPP}} = \varepsilon \cdot 4 \left(-\frac{A_5 + A_3}{\sqrt{2}} \right) = -\frac{4\varepsilon(A_5 + A_3)}{\sqrt{2}}. \quad (11)$$

With $\varepsilon \sim 10^{-26}$ and $A_5 + A_3 \sim \varphi^{-3}(1 + \varphi^{-1}) \sim 0.38$ (from the conjecture of §6), this gives $|\delta S_{\text{CPP}}| \sim 10^{-26}$, consistent with SD#1.

The angular scan at the H_4 -special angles is more sensitive: at $\theta = 36$,

$$f_{H_4}(36) = A_5 \cos(180) + A_3 \cos(108) = -A_5 + A_3 \frac{\sqrt{5} - 1}{4} \approx -A_5(1 - \varphi^{-4}), \quad (12)$$

which is larger in magnitude than $f_{H_4}(45)$ by a factor $\approx (A_5 + A_5/\varphi^4)/(A_5/\sqrt{2} + A_3/\sqrt{2}) \approx \sqrt{2}$, a modest but measurable enhancement.

6 The Amplitude Conjecture

The Fourier decomposition (7) is an exact result. The amplitudes $A_5, A_3, \dots$ require the full Nexus correlation function (SD#4). This section states the leading conjecture.

6.1 The φ^{-3} conjecture for A_5

The CPP electroweak series derived the geometric dilution factor $\varphi^{-3} \approx 0.236$ as the ratio $V_{\text{subgraph}}/V_{600\text{-cell}}$ from the $1 : \varphi : \varphi^2$ shell-radius scaling of the 600-cell (Abshier and Grok, 2026). This factor is *derived*, not fitted.

Open Problem 6.1 (Amplitude A_5 from the 600-cell geometry). *Is $A_5 = \varphi^{-3}/(2\pi)$? The conjecture rests on the observation that φ^{-3} is the universal geometric suppression factor of the 600-cell (appearing in EW boson masses, the holographic dilution, and the geometric projection of the 600-cell onto 3D subgraphs). If $A_5 = \varphi^{-3}/(2\pi)$, the amplitude of the Bell angular deviation would be determined entirely by the same geometric factor as the EW masses — a unification of the strong, EW, and quantum-foundations corrections in CPP under a single 600-cell lattice constant. Proving or disproving this conjecture requires the full Nexus derivation of SD#4.*

6.2 The A_3/A_5 ratio

The relative amplitude of the 3-fold and 5-fold components is set by the relative weight of the tetrahedral cell structure versus the icosahedral vertex structure in the Nexus constraint.

Open Problem 6.2 (Amplitude ratio A_3/A_5). *The conjectured ratio is $A_3/A_5 = \varphi^{-1} \approx 0.618$, from the golden-ratio weight of 3-fold versus 5-fold symmetry in the 600-cell. If correct:*

$$A_5 \approx \frac{\varphi^{-3}}{2\pi} \approx 0.0376, \quad (13)$$

$$A_3 \approx \frac{\varphi^{-4}}{2\pi} \approx 0.0232. \quad (14)$$

The condition for 45 to be an extremum is $5A_5 = 3A_3$, i.e., $A_3/A_5 = 5/3 \approx 1.667$. The conjecture gives $A_3/A_5 \approx 0.618 \neq 5/3$, confirming Corollary 5.2 that 45 is not an extremum for these amplitudes.

7 The Experimental Programme

7.1 What a full angular scan would show

With the Fourier decomposition (7) established, a precision measurement of $E(\theta)$ over $\theta \in [0, 180]$ would show, if CPP is correct, a residual:

$$\delta E(\theta) \equiv E_{\text{meas}}(\theta) - (-\cos \theta) = \varepsilon [A_5 \cos(5\theta) + A_3 \cos(3\theta) + \dots]. \quad (15)$$

Prediction 7.1 (The 600-cell angular signature). *A precision Bell-test scan of $E(\theta)$ over $\theta \in [0, 180]$ with angular resolution $\delta\theta < 5$ and sensitivity $\delta E < \varepsilon \cdot A_5$ will show a residual $\delta E(\theta)$ with the following properties:*

1. **Periodicity:** *The dominant period is $72 = 2\pi/5$ (from $\cos(5\theta)$), with a secondary period of $120 = 2\pi/3$ (from $\cos(3\theta)$).*
2. **Extrema at H_4 -special angles:** *Local maxima and minima of $|\delta E(\theta)|$ are located at $\theta \in \{36, 60, 72, 108, 120, 144\}$ (Table 1).*
3. **No extremum at 45:** *The point $\theta = 45$ is not a local extremum of $\delta E(\theta)$. The CHSH protocol, which measures only at $\theta = 45$, misses the dominant signal.*
4. **The distinctive cross-check:** *$\delta E(\theta)$ has a ratio of values at H_4 -special angles:*

$$\frac{\delta E(36)}{\delta E(72)} = \frac{-A_5 + A_3(\sqrt{5} - 1)/4}{A_5 + A_3(\sqrt{5} - 1)/4 - A_3} \quad (16)$$

which depends only on A_3/A_5 , not on ε . This ratio can be measured even if the absolute amplitude is below noise, by comparing the deviation at two H_4 -special angles.

If the residual $\delta E(\theta)$ does not have the 72° periodicity or has extrema at non- H_4 -special angles, CPP is falsified.

7.2 Why standard Bell tests are blind

Table 3 shows the value of f_{H_4} at the four standard CHSH angles alongside the H_4 -special angles. The CHSH angles are consistently away from the extrema, reducing the apparent signal by a factor $\sim 1/\sqrt{2}$ relative to the peak.

Table 3: Value of $f_{H_4}(\theta)/A_5$ at CHSH standard angles (bold) versus H_4 -special angles (plain), using the conjectured $A_3/A_5 = 1/\varphi$. The CHSH angles are never at extrema.

θ	$\cos(5\theta)$	$\cos(3\theta)$	f_{H_4}/A_5 (conjectured)
36	-1.000	-0.309	-1.191 $\leftarrow$ extremum
45	-0.707	-0.707	-1.144
60	0.500	-1.000	0.118 $\leftarrow$ extremum
72	1.000	-0.809	0.500 $\leftarrow$ extremum
90	0.000	0.000	0.000
108	-1.000	0.809	0.500 $\leftarrow$ extremum
120	0.500	1.000	1.118 $\leftarrow$ extremum
135	0.707	0.707	1.144
144	1.000	0.309	1.191 $\leftarrow$ extremum

Remark 7.2 (90° is exactly zero). *At $\theta = 90$: $\cos(5 \times 90) = \cos(450) = 0$ and $\cos(3 \times 90) = \cos(270) = 0$. Therefore $f_{H_4}(90) = 0$ exactly for the leading two terms. The correction vanishes at 90 regardless of the amplitudes A_5 and A_3 . This provides a stringent null test: a measurement at $\theta = 90$ should show no deviation from standard QM at leading order in CPP. Any detection of a residual at 90 would rule out the leading Fourier structure and require either higher harmonics or a modification of the H_4 symmetry argument.*

7.3 The ratio test: amplitude-independent detection

The most powerful test of the CPP angular structure is the ratio test from Prediction 7.1(4). Comparing δE at two H_4 -special angles eliminates the overall scale $\varepsilon \cdot A_5$, leaving a prediction that depends only on A_3/A_5 .

If the conjecture $A_3/A_5 = 1/\varphi$ is correct:

$$\frac{\delta E(36)}{\delta E(120)} = \frac{-A_5 - A_3 \cdot 0.309}{A_5 \cdot 0.5 + A_3} = \frac{-(1 + 0.309/\varphi)}{0.5 + 1/\varphi} \approx \frac{-1.191}{1.118} \approx -1.065. \quad (17)$$

This ratio is a prediction of CPP that does not depend on ε or on the absolute magnitude of the correction. It depends only on φ — a geometric constant of the 600-cell that is itself derived. If an experiment detects any angular structure in $\delta E(\theta)$ consistent with CPP, the ratio test (17) provides the crucial confirmation that the structure is icosahedral and not an artifact.

8 Consolidated Open Problems

Open Problem 8.1 (Derive A_5 from the Nexus). *(Restating Open Problem 6.1.) Prove or disprove $A_5 = \varphi^{-3}/(2\pi)$ from the Nexus correlation function derivation (SD#4). If true, the leading Bell deviation amplitude is the same geometric constant that governs the electroweak boson masses.*

Open Problem 8.2 (Derive A_3/A_5 from the cell-vertex weight ratio). (*Restating Open Problem 6.2.*) *The conjecture $A_3/A_5 = \varphi^{-1}$ must be derived from the relative weight of tetrahedral cell versus icosahedral vertex contributions to the Nexus constraint. This ratio is the key discriminating prediction between CPP and other hidden-variable theories that might coincidentally produce 5-fold structure.*

Open Problem 8.3 (Higher harmonics: A_{10}, A_{15}). *The suppression of A_{10} and A_{15} relative to A_5 is conjectured to be φ^{-3} per harmonic order (same as the holographic dilution hierarchy). This would give $A_{10}/A_5 \sim \varphi^{-3} \approx 0.236$ and $A_{15}/A_5 \sim \varphi^{-6} \approx 0.056$. At current sensitivity levels this is irrelevant, but it becomes important if the leading harmonics are detected.*

9 Position in the SD Series

This paper provides the angular structure of the CPP Bell correction. Together with SD#1 (framework and magnitude) and SD#3 (apparatus model and experimental architecture, forthcoming), it completes the *experimental prediction* of CPP quantum foundations. SD#4 will attempt to derive the amplitudes from first principles.

The key advance over SD#1 is specificity. SD#1 predicted a correction with “icosahedral structure”; SD#2 proves the exact Fourier decomposition, identifies the extremal angles, establishes the CHSH blind spot, and provides the amplitude-independent ratio test (17). These are falsifiable predictions that can be tested as soon as experimental precision reaches $\varepsilon \sim 10^{-26}$ in a full angular scan, or by any anomalous angular structure in near-term quantum processor Bell tests.

10 Conclusion

The H_4 symmetry of the 600-cell completely determines the angular structure of the CPP correction to Bell correlations. The main results are:

1. The projected symmetry on the measurement-angle circle is D_5 (Theorem 3.1), forcing the leading Fourier term to be $A_5 \cos(5\theta)$.
2. The independent 3-fold symmetry from tetrahedral cells adds $A_3 \cos(3\theta)$ (Theorem 4.1).
3. Extrema of f_{H_4} occur at H_4 -special angles $\{36, 60, 72, 108, 120, 144\}$ (Theorem 5.1).
4. The CHSH-optimal angle 45 is not H_4 -special (Lemma 2.1) and is not an extremum of f_{H_4} (Corollary 5.2). Standard CHSH protocols are systematically blind to the strongest CPP signal.
5. The correction vanishes exactly at $\theta = 90$ for the leading two Fourier terms — a null test prediction.
6. The ratio $\delta E(36)/\delta E(120) \approx -1.065$ (Eq. 17) is an amplitude-independent prediction that depends only on φ and can confirm icosahedral structure even below the amplitude detection threshold.

The experimental programme is clear: scan $E(\theta)$ over the full $[0, 180]$ range with angular resolution < 5 . Do not rely on the four CHSH standard angles. Measure the ratio at H_4 -special angles. Check the null result at 90.

Supplementary material: All angular calculations at
CPP/series_foundations/compute_h4_angles.py.

References

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Acknowledgements

The CPP programme is registered at OSF (DOI: <https://doi.org/10.17605/OSF.IO/JXE8D>) and maintained at GitHub (<https://github.com/Hyperphysics-Institute/Cpp>).